Vanishing of the point-to-point LFPP exponent in sufficiently high dimension

Endpoint-preserving path refinements and a Gaussian change of measure

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1. Statement and normalization

Fix $\psi > 0$. For a smooth, nonnegative, radially symmetric, compactly supported function K on $\mathbb{R}^d$ with $\int K^2 = 1$, let

$$h_n^K(x) = \int_{2^{-n}}^1 \int_{\mathbb{R}^d} K((y-x)/t) t^{-(d+1)/2}\, W(dy, dt).$$

Write $U = (-2, 2)^d$, and define

$$D_n^K(x, y; U) = \inf_{P: x \to y,\ P \subset U} \int_P e^{\psi h_n^K(z)}\, |dz|, \qquad \lambda_n^K = \operatorname{Med} D_n^K(0, e_1; U).$$

The point-to-point exponent is normalized by

$$\lambda_n^K = 2^{-n(1-\psi Q_d(\psi)) + o(n)}. \tag{1}$$

Existence of this exponent is Proposition 1.1 of Ding–Gwynne–Zhuang, *Tightness of exponential metrics for log-correlated Gaussian fields in arbitrary dimension*, arXiv:2310.03996. We will also explain why changing the regularization does not change the exponent relevant here.

Theorem. For every $\psi > 0$, there is a finite integer $d_0(\psi)$ such that, for every $d \geq d_0(\psi)$,

$$Q_d(\psi) = 0. \tag{2}$$

Equivalently, for every kernel K as above,

$$\lambda_n^K = 2^{-n + o(n)}. \tag{3}$$

The same exponent holds for smooth compactly supported mollifications of the canonical log-correlated Gaussian field, with any fixed choice of its additive constant.

The construction below uses the local path-routing idea from Lemma 2.4 of Ding–Gwynne–Zhuang, *Percolation of thick points of the log-correlated Gaussian field in high dimensions*, arXiv:2410.22075. We keep both endpoints fixed at every refinement. We do not use an implication from a set-to-set exponent to a point-to-point exponent.

2. The ball white-noise field

During the construction, use the normalized ball kernel. Put $v_d = |B_1(0)|$, where balls in this note are Euclidean unless otherwise specified, and let

$$h_n(x) = \int_{2^{-n}}^1 \int_{B_t(x)} (v_d t^d)^{-1/2} t^{-1/2}\, W(dy, dt). \tag{4}$$

Each h_n has a continuous version. For this field, $\operatorname{Var} h_n(x) = n \log 2$. Let D_n denote its internal LFPP distance in U.

We will prove the following stronger intermediate statement, without assuming existence of an exponent for this nonsmooth kernel: for sufficiently large d, there are $p_d > 0$ and deterministic $u_N = o(N)$ such that

$$\mathbb{P}\Big[D_{3N}(0, e_1; U) \le 8^{-N} e^{u_N}\Big] \ge p_d \quad \text{for all sufficiently large } N. \tag{5}$$

We use two elementary ball-overlap estimates. Define

$$R_d(s) = \frac{|B_1(0) \cap B_1(se_1)|}{v_d}.$$

There are universal $c, C > 0$ such that

$$1 - R_d(s) \le C \min\{1, \sqrt{d}\, s\}, \qquad R_d(s) \le Ce^{-cds^2}. \tag{6}$$

The second bound is understood with $R_d(s) = 0$ for $s \ge 2$. Also, for each fixed $A < \infty$, there is $c_A > 0$, independent of all sufficiently large d, such that

$$R_d(s) \ge c_A \qquad (0 \le s \le A/\sqrt{d}). \tag{7}$$

For completeness, the spherical-cap formula is

$$R_d(s) = 2\frac{v_{d-1}}{v_d} \int_{s/2}^{1} (1 - u^2)^{(d-1)/2}\, du, \quad 0 \le s \le 2.$$

Since $v_{d-1}/v_d \asymp \sqrt{d}$, integrating over $[0, s/2]$ gives the first estimate in (6). Bounding the integrand by a Gaussian gives its second estimate. Integrating over an interval $[A/(2\sqrt{d}), (A/2+1)/\sqrt{d}]$ gives (7). These estimates also follow from Lemma 3.3 of the percolation paper.

3. A family of paths with fixed endpoints

Set

$$r = \lfloor \sqrt{d} \rfloor, \qquad a_j = r^{-1} 8^{-j}, \qquad q = \lfloor d/10 \rfloor.$$

We assume throughout that d is sufficiently large that $r \ge 100$ and $q > 100$. A lattice path means a nearest-neighbor self-avoiding path; its continuous image joins consecutive vertices by straight segments.

3.1. Local routing

For neighboring vertices x, y of the lattice $a_j\mathbb{Z}^d$, consider the fine-lattice tube

$$T_{x,y} = \{z \in a_{j+1}\mathbb{Z}^d : \operatorname{dist}_1(z, [x, y]) \le 2a_{j+1}\}.$$

There are q mutually vertex-disjoint paths in this tube, each having ten edges, with its first vertex at distance a_{j+1} from x and its last vertex at distance a_{j+1} from y. If the first vertex z, with $|z - x|_1 = a_{j+1}$, is prescribed instead, there are q such paths that intersect one another only at z. All these paths avoid the coarse vertices x, y.

Here is an explicit construction, after scaling and translating so that $x = 0$, $y = 8u$, and u is a signed coordinate vector. Choose mutually disjoint auxiliary coordinate sets, avoiding the coordinate of u and, when relevant, that of z.

For a free start, the route associated with auxiliary coordinates k, l is

$$e_k,\ u + e_k, \dots, 8u + e_k,\ 8u + e_k + e_l,\ 8u + e_l.$$

For a prescribed start $z = u$, omit the first point e_k and start at u:

$$u,\ u + e_k,\ 2u + e_k, \dots, 8u + e_k,\ 8u + e_k + e_l,\ 8u + e_l.$$

For $z = -u$, use

$$-u,\ -u + e_k,\ e_k,\ u + e_k, \dots, 8u + e_k.$$

For z perpendicular to u, use

$$z,\ z + e_k,\ z + u + e_k, \dots, z + 8u + e_k,\ 8u + e_k.$$

Each displayed route has ten edges, stays in the indicated tube, and avoids $0, 8u$. Distinct auxiliary coordinate sets give the asserted disjointness. There are enough coordinates to choose q routes.

3.2. Pinned refinement

Let $A = 0$, $B = e_1$, and let P_0 be the straight lattice path from A to B on $a_0\mathbb{Z}^d$. Given a path P_j, refine its coarse edges chronologically using the preceding routes. For the first edge, use a free-start family. For every later edge, prescribe its start to be the last vertex of the previous refined edge.

At each choice, discard routes that intersect the previous piece except at their shared endpoint. At most twelve routes are discarded: distinct candidate routes are disjoint away from their prescribed start, and the preceding piece has at most twelve vertices. Choose a deterministic subset of $q - 20$ remaining routes and choose uniformly from it.

Finally add the edge from A to the first free vertex, and the edge from the last free vertex to B. The original routes avoid the coarse endpoints. The added edges therefore preserve self-avoidance. Indeed, nonincident coarse edges are at distance at least a_j, whereas the two tube radii sum to $a_j/2$; and a coarse endpoint is at distance at least a_j from any nonincident coarse edge. Consecutive pieces were made disjoint by the exclusions.

We obtain a probability law ν_N on finite refinement histories $(P_0, \dots, P_N)$. There is no need to identify histories that happen to have the same final geometric path.

Assign each added endpoint edge to the first or last coarse parent edge. Every fine edge then has a specified parent edge. The number of edges satisfies

$$E_{j+1} = 10E_j + 2, \qquad E_0 = r. \tag{8}$$

Every one-step parent piece has at most twelve vertices, and every parent edge has at most twelve descendant edges per subsequent generation. The first and last parent pieces are distinct because $E_j \geq r \geq 100$.

All descendants of an edge at level i stay within ℓ^1-distance

$$\sum_{j>i} 2a_j = \frac{2}{7}a_i \tag{9}$$

of that edge. In particular, every point of P_N is within $2a_i$ of a vertex of P_i, for each $i \leq N$. All the paths lie in a fixed $2a_0/7$-neighborhood of $[0, e_1]$, hence in U.

3.3. Uniform local occupancy

There is a universal constant C such that every path in this construction satisfies

$$\#(P_j \cap B(x, Ra_j)) \le C(1+R)^2 \quad (j \ge 0,\ x \in \mathbb{R}^d,\ R \ge 0). \tag{10}$$

To prove this for $R \ge 1$, choose $s = \lceil \log_8(C_0 R) \rceil$, where C_0 is an absolute constant large enough that

$$R8^{-s} + \frac{2}{7} < \frac{1}{2}.$$

If $s \le j$, each ancestor edge at level $j-s$ with descendants in the ball lies at ℓ^∞-distance less than $a_{j-s}/2$ from its center. Two nonincident nearest-neighbor lattice edges have ℓ^∞-distance at least a_{j-s}. Thus any two relevant ancestor edges must be incident. A self-avoiding path in a cubic lattice has at most two pairwise incident edges. There are consequently at most two relevant ancestor edges, each having at most $C12^s$ descendant vertices. This is at most $C(1+R)^2$, since $\log_8 12 < 2$.

If $s > j$, use the straight root path. At most $C(1+R8^{-j})$ root edges are relevant. Their descendants number at most

$$C12^j(1+R8^{-j}) \le C(1+R)^2,$$

using $8^j \le CR$. The case $R < 1$ follows from the case $R = 1$.

In particular, writing $\mathcal{L}$ for Euclidean length, for each fixed R,

$$\mathcal{L}(P_N \cap B(A, Ra_m)) + \mathcal{L}(P_N \cap B(B, Ra_m)) \le C_R a_N 12^{N-m}, \quad 1 \le m \le N, \tag{11}$$

and

$$\mathcal{L}(P_N) \le C_d a_N 12^N. \tag{12}$$

For (11), trace each fine edge meeting the ball to level m, use (9) to enlarge the ball by a fixed multiple of a_m, and use (10) to bound the number of ancestor edges. Counting entire fine edges only increases the length.

4. Intersections form a subcritical forest

Let P, Q be two independent histories with law ν_N. Their common vertices admit a domination that retains both genealogy and spatial location.

Intersection lemma. Put

$$p = \frac{60}{q-20}.$$

For sufficiently large d, the common vertices can be injected into a branching forest with at most $r+1$ initial particles, two new immigrants at each subsequent generation, and independent offspring distribution

$$X = 12\ \mathrm{Bin}(2, p). \tag{13}$$

The two immigrants at each generation represent A, B. Every nonimmigrant actual common child at level $j+1$ has an actual common parent at level j, at distance at most $5a_j/4$.

Proof. Conditional on both histories and the existing forest embedding through level j, first reveal Q_{j+1}, then reveal the edge refinements making P_{j+1} in chronological order.

A coarse edge of P_j disjoint from Q_j cannot produce an intersection with Q_{j+1}, by tube separation. Charge each eligible coarse edge to one of its endpoints in $P_j \cap Q_j$, choosing deterministically if there are two. Each common parent receives at most two eligible edges.

Within an eligible tube, Q_{j+1} occupies at most $4 \cdot 12 \le 48$ vertices. To see this, a coarse edge of Q_j whose refinement enters that tube must intersect the eligible coarse edge; there are at most four such edges, because Q_j is self-avoiding.

For the first piece, call its trial successful if the pinned piece meets Q_{j+1} anywhere other than A. In particular, an intersection at its freely chosen first fine vertex counts as success. For each later piece, omit its prescribed fine starting vertex from the success test; for the final piece also omit B. Candidate pieces are mutually disjoint away from these forced vertices. At most sixty of the candidates can therefore cause success. Conditional on all preceding revelations, the success probability is at most p.

Every actual common vertex other than A, B belongs to a successful piece, possibly after assigning a shared start to the preceding piece whose end it is. Each successful piece accounts for at most twelve common vertices. Assign them to the common parent charged by that coarse edge. Their distances to that parent are at most

$$a_j + 2a_{j+1} = \frac{5}{4} a_j.$$

For completeness, the branching domination is a sequential coupling, not merely a bound for the total intersection count. For each actual parent allocate two trial slots; the parent and slot of each eligible edge are fixed before its fresh uniform variable is sampled. At each trial its conditional success probability is some $p' \le p$; realize its indicator using a fresh uniform variable as $\mathbf{1}_{\{V \le p'\}}$, and dominate it by $\mathbf{1}_{\{V \le p\}}$. Sample the remaining choice of the refinement conditionally on its success indicator, preserving its original uniform law. Put the at most twelve actual children into the slots of the corresponding successful dominating trial. Pad missing slots and all unused descendants with independent dummy offspring. Iterating this construction gives the independent forest (13), together with the required injection and actual parent-child locations. The forest need not be independent of the paths; its offspring variables have the required independent product law. The endpoints are inserted as new immigrants at every generation. □

A descendant at any later level of an endpoint immigrant born at level b stays within distance

$$\frac{5}{4} \sum_{j=b}^{\infty} a_j = \frac{10}{7} a_b \tag{14}$$

of that endpoint. Therefore a common vertex at level i outside endpoint balls of radius $(10/7)8^H a_i$ either descends from an initial particle or belongs to an immigrant family of age at least H. No spatial locations need be assigned to dummy forest particles.

4.1. Exponential moments of old families

Fix $\theta > 0$. For all sufficiently small $p = p(\theta)$, there is $F_\theta < \infty$ such that, for the total progeny T of one particle,

$$\mathbb{E} e^{\theta T} \le F_\theta,$$

and, for the total number $T_{\ge H}$ of its descendants of ages at least H,

$$\mathbb{E} e^{\theta T_{\ge H}} \le 1 + C_\theta 2^{-H}. \tag{15}$$

Indeed, let $g(s) = (1 - p + ps^{12})^2$, and choose $R = 2e^\theta$. For small enough p,

$$e^\theta g(R) \le R, \qquad \sup_{1 \le s \le R} g'(s) \le \frac{1}{2}.$$

The recursion for total progeny truncated at generation m gives an exponential moment bounded by R, uniformly in m. Monotone convergence gives $F = \mathbb{E}e^{\theta T} \le R$. Conditioning on generation H gives

$$\mathbb{E}e^{\theta T_{\ge H}} = g^{\circ H}(F) \le 1 + (R-1)2^{-H}.$$

Independence of the forest trees now implies that, if Z_i^{old} counts the initial families and immigrant particles of age at least H at level i, then

$$\mathbb{E}\exp\left(\theta \sum_{i=0}^{N} Z_i^{\mathrm{old}}\right) \le \exp\Big(C_\theta(r+1) + C_\theta(N+1)2^{-H}\Big). \tag{16}$$

The full progeny of the initial particles was included; it only contributes a constant depending on d, θ.

5. A dimension-uniform bound for pruned tube overlap

The white-noise correlation concerns nearby paths, not only exact common vertices. We now supply the required comparison, with a constant independent of dimension.

Let $\ell \ge 1$, $R = 8^\ell$, $E = \{A, B\}$, and retain the vertices

$$S_j(P) = \{x \in P_j : \mathrm{dist}(x, E) > Ra_j\}.$$

For $1 \le j \le N$, define

$$f_{P,j}(y,t) = \mathbf{1}_{\{8^{-j} \le t \le 2\cdot 8^{-j}\}} \mathbf{1}_{\{y \in B_t(S_j(P))\}} (v_d t^d)^{-1/2} t^{-1/2}, \qquad f_P = \sum_{j=1}^{N} f_{P,j}. \tag{17}$$

The time bands are disjoint. All these functions belong to the white-noise Hilbert space $L^2(dy\,dt)$.

Pruned-overlap lemma. There are universal constants C_*, C_0 and integers

$$J_d = \lceil \log_8(C_0 r) \rceil + 2$$

such that, whenever $\ell \ge J_d + C_0$,

$$\langle f_P, f_Q \rangle \le C_* \sum_{i=0}^{N} \#\left\{ v \in P_i \cap Q_i : \mathrm{dist}(v, E) > \frac{8^\ell}{C_0\, 8^{J_d}}\, a_i \right\}. \tag{18}$$

Proof. By (6), for $t \in [8^{-j}, 2\cdot 8^{-j}]$,

$$\frac{|B_t(x) \cap B_t(y)|}{v_d t^d} \le C \exp\left(-c\frac{|x-y|^2}{a_j^2}\right) \mathbf{1}_{\{|x-y| < 4ra_j\}}. \tag{19}$$

All constants here are universal, since $r \le \sqrt{d}$. The integral dt/t of one time band is $\log 2$.

The occupancy bound (10), divided into dyadic annuli, gives both

$$\sum_{y \in Q_j} e^{-c|x-y|^2/a_j^2} \le C \tag{20}$$

and, for $L \ge 1$,

$$\sum_{\substack{y \in Q_j \\ |x-y| \ge La_j}} e^{-c|x-y|^2/a_j^2} \le Ce^{-c'L^2}. \tag{21}$$

We may enlarge the second tube in the overlap to the full tube of Q_j.

First consider active vertices $x \in S_j(P)$ incident to an edge of P_j that meets Q_j. Charge such a vertex to a common endpoint of that edge. This common vertex is within distance a_j of x, and each common vertex receives at most four charges. Equations (19)–(20) bound all these contributions by a universal constant times the common-vertex count outside the slightly smaller endpoint collar.

For every remaining active vertex x, choose an incident edge e_j of P_j. It is disjoint from Q_j. Follow its specified edge ancestry backward. Since $P_0 = Q_0$, there is a first level s such that the ancestor edge e_s is disjoint from Q_s, but its parent e_{s-1} intersects Q_{s-1}.

After this separation, descendants stay separated. At level s, the Euclidean distance between e_s and the continuous path Q_s is at least a_s. Equation (9) for the two paths gives, at every later level j,

$$\operatorname{dist}(\text{descendants of } e_s,\ Q_j) \geq \frac{3}{7} a_s. \tag{22}$$

The subtree of e_s has at most $C12^{j-s}$ vertices at level j. By (21) for $j > s$, and by (20) for $j = s$, its total pair-kernel contribution at level j is at most

$$C12^{j-s} \exp(-c' 8^{2(j-s)}).$$

Summing over all $j \geq s$ gives a universal constant. This summation, rather than a separate estimate for each of the $O(\log d)$ generations, is what makes the coefficient in (18) independent of d.

Include a separating edge in the charged collection only when some active vertex x at a later level j witnesses a nonzero contribution from its subtree. Once an edge is included, its entire descendant contribution is bounded by the preceding universal sum. For its active witness, the support indicator in (19) and (22) imply

$$8^{j-s} \leq Cr, \qquad j - s \leq J_d. \tag{23}$$

Charge the separating edge e_s to a common endpoint v of its parent e_{s-1}. The descendant displacement bound gives

$$|x - v| \leq \frac{9}{7} a_{s-1}.$$

Consequently,

$$\operatorname{dist}(v, E) > \left(\frac{R}{8^{J_d+1}} - \frac{9}{7} \right) a_{s-1} \geq \frac{R}{C_0\, 8^{J_d}} a_{s-1}, \tag{24}$$

after increasing C_0 and imposing the stated lower bound on ℓ.

Each common vertex is incident to at most two parent edges, and each parent has at most twelve fine child edges. Thus at most twenty-four separating subtrees are charged to a common vertex. Combining their universally bounded contributions with the direct charges proves (18). □

Choose a universal integer C_1 sufficiently large and set

$$\ell_N = \lceil \sqrt{N} \rceil + 2J_d + C_1. \tag{25}$$

By (14), every actual common vertex counted on the right side of (18) belongs to an initial family or an immigrant family of age at least $H_N = \lfloor \sqrt{N} \rfloor$. Apply (16) with $\theta = C_* \beta^2$. For every fixed $\beta > 0$, once d is sufficiently large depending only on β,

$$\sup_{N \geq 1} \mathbb{E}_{P,Q} \exp\Big(\beta^2 \langle f_P, f_Q \rangle \Big) \leq C_{d,\beta} < \infty. \tag{26}$$

Changing $C_{d,\beta}$ covers the finitely many small N. The quantity $(N+1)2^{-\lfloor \sqrt{N} \rfloor}$ is bounded. Crucially, d is chosen before N, and before any limiting argument in the collar size.

6. Cheap fixed-endpoint paths under a Gaussian tilt

For a history P, let

$$L_P = \exp\left(-\beta W(f_P) - \frac{\beta^2}{2}\|f_P\|_2^2\right). \tag{27}$$

Its expectation is one. Under the probability measure with density L_P, the noise $W^0 = W + \beta f_P$ is standard white noise. If G_m is the centered field constructed from W^0, then

$$h_{3N}(z) = G_{3N}(z) + \mu_P(z), \qquad \mu_P(z) = -\beta\langle g_{N,z}, f_P\rangle, \tag{28}$$

where $g_{N,z}$ is the kernel in (4) with lower cutoff 8^{-N}.

Put $\rho(z) = \min\{|z-A|, |z-B|\}$. Partition the path into shells

$$\mathcal{S}_0 = \{\rho > a_1\}, \qquad \mathcal{S}_m = \{a_{m+1} < \rho \le a_m\}\ (1 \le m < N), \qquad \mathcal{S}_N = \{\rho \le a_N\}.$$

Assign each point to a nearest endpoint, resolving ties deterministically.

If $z \in \mathcal{S}_m$, $m < N$, and $i \ge m + \ell_N + 2$, some ancestor vertex $v \in P_i$ lies within $2a_i$ of z. Moreover

$$\operatorname{dist}(v, E) \ge \rho(z) - 2a_i > 8^{\ell_N} a_i,$$

so $v \in S_i(P)$. For $t \in [8^{-i}, 2\cdot 8^{-i}]$, the ratio $|z-v|/t$ is at most $2/r \le 4/\sqrt{d}$. Equations (7) and (17) yield a universal $c_* > 0$ such that

$$\mu_P(z) \le -\beta c_* \log 2\,(N - m - \ell_N - 2)_+. \tag{29}$$

For $m = N$, the same bound states only $\mu_P \le 0$, which follows from nonnegativity of the kernels.

We estimate shell lengths by subtracting the centered coarse field at the nearby deterministic endpoint. If $|z-x| \le a_m$, $x \in \{A, B\}$, then the independent scale bands and (6) give

$$\begin{aligned}\operatorname{Var}(G_{3N}(z) - G_{3m}(x)) &= 3(N-m)\log 2 + 2\int_{8^{-m}}^{1} [1 - R_d(|z-x|/t)]\,\frac{dt}{t} \\ &\le 3(N-m)\log 2 + C,\end{aligned} \tag{30}$$

with C independent of d, N, m, z: use $\sqrt{d}\,|z-x| \le 2\cdot 8^{-m}$ in the integral.

Let $\mathbb{E}_P$ denote expectation under the tilted measure. For $m \ge 1$, define

$$Y_{m,x} = e^{-\psi G_{3m}(x)} \int_{P_N \cap \mathcal{S}_m \text{ assigned to } x} e^{\psi h_{3N}(z)}\,|dz|.$$

For $m = 0$, use the same integral without a coarse-field factor. By Tonelli's theorem, the Gaussian exponential-moment identity, and (11)–(12), (29)–(30),

$$\mathbb{E}_P Y_{m,x} \le C_d a_N 12^{N-m} \exp\left[-\psi\beta c_* \log 2\,(N - m - \ell_N - 2)_+ + \frac{3}{2}\psi^2 (N-m)\log 2 + C\psi^2\right]. \tag{31}$$

For $m = 0$, use $\operatorname{Var} G_{3N}(z) = 3N\log 2$ and the total-length bound; the same estimate holds.

Choose $\beta = \beta(\psi)$, independent of d, so that

$$\psi\beta c_* \log 2 > \log 12 + \frac{3}{2}\psi^2 \log 2 + 1. \tag{32}$$

The sum in (31), with $q = N - m$, is a geometric series after $q > \ell_N + 2$. Hence, uniformly over every history P,

$$\mathbb{E}_P \sum_{m,x} Y_{m,x} \le C_{d,\psi} a_N e^{C_\psi \ell_N}. \tag{33}$$

Let $t_N = N^{3/4}$. Markov's inequality gives

$$\mathbb{P}_P \left[\sum_{m,x} Y_{m,x} > a_N e^{C_\psi \ell_N + t_N} \right] \longrightarrow 0.$$

The centered endpoint fields have variance at most $3N \log 2$. A union bound gives

$$\mathbb{P}_P \left[\max_{\substack{0 \le m \le N \\ x \in \{A,B\}}} G_{3m}(x) > t_N \right] \le 2(N+1) \exp\left(-\frac{N^{1/2}}{6 \log 2} \right) \longrightarrow 0. \tag{34}$$

These estimates are uniform in P. On their complementary events,

$$\int_{P_N} e^{\psi h_{3N}(z)} \, |dz| \le a_N \exp(C_\psi \ell_N + (1+\psi) t_N) \le 8^{-N} e^{u_N}, \tag{35}$$

where $u_N = C_\psi \ell_N + (1+\psi) N^{3/4} = o(N)$.

The centered field G, rather than the original field h, was subtracted in the definition of $Y_{m,x}$. Thus (30)–(31) use the exact Gaussian covariance under the tilted law; no independence between an endpoint field value and a shell integral was assumed.

7. Returning to the original field

Define the nonnegative random variable

$$X_N(W) = \mathbb{E}_{P \sim \nu_N} \left[L_P(W) \, \mathbf{1}_{\{ \mathrm{len}_{h_{3N}}(P_N) \le 8^{-N} e^{u_N} \}} \right]. \tag{36}$$

Uniformity of (35) gives

$$\mathbb{E} X_N = 1 - o(1).$$

For any pair of histories, the exact Gaussian calculation is

$$\mathbb{E}[L_P L_Q] = \exp(\beta^2 \langle f_P, f_Q \rangle). \tag{37}$$

Choose d sufficiently large for (26), with the $\beta(\psi)$ fixed in (32). Dropping the two length indicators and using (26),

$$\mathbb{E} X_N^2 \le C_{d,\psi}.$$

Cauchy–Schwarz therefore yields

$$\mathbb{P}[X_N > 0] \ge \frac{(\mathbb{E} X_N)^2}{\mathbb{E} X_N^2} \ge \frac{1}{2 C_{d,\psi}}$$

for all sufficiently large N. Whenever $X_N > 0$, at least one finite-history polygonal path from 0 to e_1, contained in U, has the length bound (35). This proves (5).

The bound need not have probability close to one. A positive probability depending on the fixed dimension is enough to identify the exponential scale, by Gaussian concentration.

8. Regularization transfer and identification of the exponent

We record the regularization comparison carefully to avoid imposing a smoothness assumption on the ball kernel.

8.1. Comparison on a common canonical field

Let φ be the canonical whole-space log-correlated Gaussian field with a fixed additive normalization, and let ρ be a smooth, nonnegative, compactly supported mollifier of integral one. Write $\rho_n(x) = 2^{nd}\rho(2^n x)$.

There is a coupling of the ball field h and φ such that $h - \varphi$ is continuous; this is Lemma 4.2 of the percolation paper. Its Lemma 4.1 gives, on every fixed compact set,

$$\|h_n - h * \rho_n\|_\infty \le n^{2/3} \quad \text{eventually almost surely,} \tag{38}$$

up to a fixed constant and an enlargement of the compact set. The exceptional probabilities are summable. Consequently,

$$\|h_n - \varphi * \rho_n\|_\infty = o(n) \quad \text{almost surely.} \tag{39}$$

For every smooth kernel K in the theorem, completing its white-noise representation over $t \in (0,\infty)$ gives the same canonical field modulo a constant; the contribution from $t \ge 1$ has a continuous version. This is Remark 1.4 of the tightness paper. The cutoff-versus-mollification comparison (38) holds for this kernel as well. One can see this directly: for

$$Z_n = h_n^K - h^K * \rho_n,$$

the white-noise isometry gives $\sup_x \mathbb{E}Z_n(x)^2 \le C_{d,K,\rho}$. Here are the estimates explicitly. Put $\varepsilon = 2^{-n}$ and $K_t(y) = t^{-d/2}K(y/t)$. For $t \ge \varepsilon$,

$$\|K_t - K_t * \rho_\varepsilon\|_2 \le C\varepsilon/t,$$

so its squared norm is integrable against dt/t. For $t < \varepsilon$, Young's inequality gives

$$\|K_t * \rho_\varepsilon\|_2 \le \|K_t\|_1 \|\rho_\varepsilon\|_2 \le C(t/\varepsilon)^{d/2},$$

whose square is again integrable against dt/t. Applying the same estimates to derivatives, using a derivative of ρ_ε below ε and a derivative of K_t above it, bounds the gradient variance of Z_n by $C\varepsilon^{-2}$. Consequently,

$$\mathbb{E}(Z_n(x) - Z_n(y))^2 \le C_{d,K,\rho}2^{2n}|x-y|^2 \le C_{d,K,\rho}2^n|x-y|, \quad |x-y| \le 2\cdot 2^{-n}.$$

Cover a compact set by $O(2^{nd})$ balls of radius 2^{-n}. Gaussian entropy bounds on each ball give a uniformly bounded expected oscillation, and Gaussian concentration plus a union bound gives

$$\mathbb{P}[\|Z_n\|_\infty > n^{2/3}] \le Ce^{-cn^{4/3}}$$

for all sufficiently large n, after changing fixed constants. This proves the analogue of (38).

Glue the two couplings over their common marginal φ. Both fields are random elements of separable distribution spaces, so regular conditional distributions give such a joint coupling. Equations (38)–(39) then imply

$$\|h_n - h_n^K\|_{\overline{U}} = o(n) \quad \text{almost surely.} \tag{40}$$

Thus, path by path and then after taking infima,

$$e^{-\psi\|h_n - h_n^K\|_{\overline{U}}} D_n \le D_n^K \le e^{\psi\|h_n - h_n^K\|_{\overline{U}}} D_n. \tag{41}$$

No metric tightness or comparison of set-to-set and point-to-point normalizations was used.

8.2. The upper bound on the point exponent

By (5), (40)–(41), for each fixed $\eta > 0$,

$$\liminf_{N\to\infty} \mathbb{P}\Big[D^K_{3N}(0, e_1; U) \le \exp(-3N \log 2 + \eta N)\Big] \ge p_d. \tag{42}$$

To justify the probability statement, intersect the positive-probability event in (5) with the comparison event in (40), whose probability tends to one. No independence is required.

The logarithm of LFPP distance satisfies Gaussian concentration with variance proxy $\psi^2 n \log 2$; see Lemma 2.7 of the tightness paper. This also follows directly from the Cameron–Martin bound

$$\sup_x |h^K_n(x; W + f) - h^K_n(x; W)| \le \sqrt{n \log 2}\, \|f\|_2,$$

which makes $\log D^K_n$ a $\psi\sqrt{n \log 2}$-Lipschitz Gaussian functional. Therefore its fixed quantiles differ by $O_{d,\psi}(\sqrt{n})$. A positive lower probability in (42) forces

$$\limsup_{N\to\infty} \frac{1}{3N} \log \lambda^K_{3N} \le -\log 2. \tag{43}$$

For example, compare the threshold in (42) to the median; a positive linear excess of the median over that threshold would make the lower-tail probability decay exponentially in N, contradicting (42). First obtain the bound $-\log 2 + \eta/3$ for each $\eta > 0$, and then let $\eta \downarrow 0$.

8.3. The endpoint lower bound

For any fixed dimension and every kernel in the theorem,

$$D^K_n(0, e_1; U) \ge 2^{-n} \exp\left(\psi \inf_{z\in B_{2^{-n}}(0)} h^K_n(z)\right). \tag{44}$$

Indeed, every connecting path has an initial portion that exits this Euclidean ball, and that portion has Euclidean length at least 2^{-n}.

The value $h^K_n(0)$ is centered Gaussian of variance $n \log 2$. The oscillation

$$\sup_{|z|\le 2^{-n}} |h^K_n(z) - h^K_n(0)|$$

is tight as $n \to \infty$, for fixed d, K. To check this, rescale the ball to unit size. The white-noise isometry and smoothness of K bound the canonical increment metric by $C_{d,K}|u - v|$, uniformly in n; Gaussian entropy and concentration give the stated tightness. Thus the infimum in (44) is $o(n)$ in probability. It follows that

$$\liminf_{n\to\infty} \frac{1}{n} \log \lambda^K_n \ge -\log 2. \tag{45}$$

In particular $Q_d(\psi) \ge 0$.

Combining (43)–(45) with existence of the exponent in (1) gives

$$-(1 - \psi Q_d(\psi)) \log 2 = -\log 2.$$

Since $\psi > 0$, this is exactly $Q_d(\psi) = 0$.

Finally, the same comparison with $\varphi * \rho_n$ proves the stated canonical-mollification version. Every dimension restriction in the construction is either universal or is the requirement that $p = 60/(\lfloor d/10 \rfloor - 20)$ be sufficiently small for the fixed value $\theta = C_* \beta(\psi)^2$. Hence one finite $d_0(\psi)$ works for every $d \ge d_0(\psi)$, as asserted. □

References

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